ATAI Evaluation Event III

TitaniumTwinklingSnake

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1 Agent Overview and Capabilities

Our Stage 3 agent is a guarded movie chatbot that combines the Stage 2 knowledge-graph (KG) question-answering system with a local open-weight language model, Qwen2.5-1.5B-Instruct. Every incoming Speakeasy message is classified before routing as `safe_in_scope`, `out_of_scope`, `toxic`, or `unsafe`. Safe factual questions are resolved against the movie graph through entity/relation matching and SPARQL. Open-ended movie questions use Qwen under a restricted movie-domain system prompt. Non-movie requests are redirected, abusive inputs receive a short neutral response, and dangerous requests are refused. The submission also supports a fully local Qwen checkpoint so inference can run on the Nuvolos CUDA GPU without downloading model files at evaluation time.

2 Adopted Methods

- **Knowledge-graph QA.** The Stage 2 component matches entities and relations, generates SPARQL, executes it with RDFLib, and verbalizes graph results. It remains the preferred path for precise factual answers.
- **Open-weight generation.** We use Qwen2.5-1.5B-Instruct with PyTorch and Hugging Face Transformers. Generation is deterministic, short, and constrained by a movie-domain prompt. The packaged checkpoint includes the model configuration, tokenizer, vocabulary, and weights for local loading.
- **Safety and scope classification.** Obvious toxic and dangerous patterns are handled by deterministic checks, while Qwen provides semantic classification for less literal inputs. Safety takes priority over prompt-injection-like instructions. Classifier or generation failures fail closed instead of silently producing unrestricted output.
- **Speakeasy integration.** The client polls assigned rooms, passes statements as well as questions to the guardrail layer, and sends the returned answer back to the conversation.

3 Pipeline

For "*Who directed Inception?*", the guardrail first labels the message `safe_in_scope`. The factual agent resolves *Inception*, maps the relation *directed by* to Wikidata property P57, executes SPARQL over the course graph, and retrieves *Christopher Nolan*. The verified graph result is then returned directly or can be supplied to Qwen for a constrained rewrite, depending on the configured path. For an open-ended request such as "*What is Inception about?*", no suitable factual graph answer is produced, so the agent falls back to Qwen. If the classifier instead returns `out_of_scope`, `toxic`, or `unsafe`, the request is answered immediately by the corresponding guarded response and is not sent to the factual QA path.

4 Improvements

The main improvement over Stage 2 is guarded routing rather than treating every message as a factual question. During live testing we found that the old client ignored non-question statements, causing the required toxic example "*You are a useless bot.*" to bypass the guardrail. Removing that filter made statements reach the classifier and fixed the failure. We also strengthened fail-closed behavior for model initialization/classification/generation errors, added prompt-injection handling, and preserved graph-backed answers when the LLM is unavailable. Finally, the Qwen checkpoint is bundled and was verified with Hugging Face offline mode on CUDA, eliminating network dependence during normal inference.

5 Additional Features

We added a 300-prompt evaluation suite plus automated guardrail unit tests. The diagnostic set contains 70 safe factual, 55 safe open-ended, 60 out-of-scope, 55 toxic, and 60 unsafe prompts. In the recorded diagnostic run, 245 cases were behavior-evaluated while 55 open-ended generation cases were marked pending rather than failed; 222/245 evaluated behaviors passed. The guardrail classifier achieved 294/300 correct labels (98%). Category behavior results were 53/70 safe factual, 60/60 out-of-scope, 50/55 toxic, and 59/60 unsafe. Ten targeted unit tests covering required examples, failure handling, mixed harmful/movie requests, and prompt-injection priority all pass. The full Nuvolos graph loads 2,413,825 triples, 123,174 labels, and 497 relation properties.

Category	**Passed**	**Evaluated**	**Pending**
Safe factual	53	70	0
Safe open-ended	–	0	55
Out-of-scope	60	60	0
Toxic	50	55	0
Unsafe	59	60	0
Evaluated overall	222	245	55